

A Mechanistic Model for Radiation-Induced Segregation Controlled Grain-Boundary Oxidation in Austenitic Stainless Steels

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1 Introduction

2 Model Formulation and Calibration

The model consists of two components: a radiation-induced segregation (RIS) model for the evolving GB chemistry and a reduced-dimensional oxidation model for oxygen transport, consumption, and oxide growth. The RIS-predicted Cr, Ni, Si, and Fe concentrations provide the local alloy composition for the oxidation model. Oxygen transport along the GB is solved explicitly, while transport through the oxide and interfacial reactions are treated using a quasi-steady resistance formulation. The oxide composition determines the effective oxygen diffusivity through an effective-medium treatment, which controls the local oxygen consumption and oxide growth rate. The formulations of the two components are described below.

Notation. Alloying-element indices are $i, j \in \{\text{Cr}, \text{Ni}, \text{Si}, \text{Fe}\}$; oxide-phase indices are $k \in \{\text{Cr}_2\text{O}_3, \text{Fe}_3\text{O}_4, \text{FeCr}_2\text{O}_4, \text{SiO}_2\}$. Superscripts V and I denote quantities associated with vacancies and self-interstitials; the superscript GB denotes a value taken on the grain boundary. C is a site fraction, $\mathbf{J}$ a flux in the two-dimensional metal domain, and J a flux on the one-dimensional grain-boundary line. The coordinate x is normal to the boundary and y runs along it.

2.1 Radiation-Induced Segregation Model

The RIS model is based on solving a set of coupled mass-conservation equations for the alloying elements (Cr, Ni, Si, and Fe) and point defects (vacancies and self-interstitials). For each alloying element i , the concentration evolves according to

$$\frac{\partial C_i}{\partial t} = -\nabla \cdot \mathbf{J}_i. \quad (1)$$

The site fractions of vacancies and self-interstitials are governed separately by

$$\frac{\partial C_V}{\partial t} = -\nabla \cdot \mathbf{J}_V + \epsilon K_0 - R C_I C_V, \quad (2)$$

$$\frac{\partial C_I}{\partial t} = -\nabla \cdot \mathbf{J}_I + \epsilon K_0 - R C_I C_V, \quad (3)$$

where C_V and C_I are the vacancy and self-interstitial site fractions, ϵ is the damage efficiency, K_0 is the radiation dose rate, and R is the vacancy-interstitial recombination rate coefficient. The flux of each alloying element carried by each defect mechanism is

$$\mathbf{J}_i^V = -\sum_j L_{ij}^V \nabla(\mu_j - \mu_V), \quad (4)$$

$$\mathbf{J}_i^I = -\sum_j L_{ij}^I \nabla(\mu_j + \mu_I), \quad (5)$$

with the Onsager coefficients

$$L_{ii}^V = \frac{C_i D_i^V C_V}{k_B T} \left(1 + \frac{1 - f_0^V}{f_0^V} \frac{C_i D_i^V}{\sum_m C_m D_m^V} \right), \quad (6)$$

$$L_{ij}^V = L_{ji}^V = \frac{1 - f_0^V}{f_0^V} \frac{C_i D_i^V C_j D_j^V C_V}{k_B T \sum_m C_m D_m^V}, \quad i \neq j, \quad (7)$$

$$L_{ii}^I = \frac{C_i D_i^I C_I}{k_B T} \left(1 + \frac{1 - f_0^I}{f_0^I} \frac{C_i D_i^I}{\sum_m C_m D_m^I} \right), \quad (8)$$

$$L_{ij}^I = L_{ji}^I = \frac{1 - f_0^I}{f_0^I} \frac{C_i D_i^I C_j D_j^I C_I}{k_B T \sum_m C_m D_m^I}, \quad i \neq j, \quad (9)$$

where D_i^V and D_i^I are the mechanism-specific diffusion coefficients of element i , and f_0^V and f_0^I are the corresponding correlation factors. The net defect

fluxes follow from conservation of lattice sites,

$$\mathbf{J}_V = - \sum_i \mathbf{J}_i^V, \quad (10)$$

$$\mathbf{J}_I = \sum_i \mathbf{J}_i^I, \quad (11)$$

and give rise to a lattice velocity

$$\mathbf{v}_{\text{lat}}^{\text{RIS}} = \mathbf{J}_V - \mathbf{J}_I. \quad (12)$$

2.2 Grain-Boundary Oxidation Model

Oxygen transport along the boundary, treated as a slab of width δ_{GB} , obeys

$$\frac{\partial C_O}{\partial t} = - \frac{\partial J_O}{\partial y} - \frac{J_r}{\delta_{\text{GB}}}, \quad (13)$$

$$J_O = -D_O^{\text{eff}} \frac{\partial C_O}{\partial y}, \quad (14)$$

where the effective diffusivity is obtained from an effective-medium mixture of the oxide phases,

$$D_O^{\text{eff}} = \sum_k f_k D_O^k, \quad (15)$$

with f_k the volume fraction and D_O^k the oxygen diffusivity of phase k . Oxygen delivery from the boundary channel to the metal/oxide interface is treated as a quasi-steady series resistance across the oxide of total thickness $\sum_k L_k$,

$$J_r = D_O^{\text{eff}} \frac{C_O - C_O^{\text{int}}}{\sum_k L_k}, \quad (16)$$

where L_k is the thickness of oxide k and C_O^{int} is the oxygen concentration at the interface. The interfacial reaction rates of the four oxide phases are

$$J_r^{\text{Cr}_2\text{O}_3} = k_r^{\text{Cr}_2\text{O}_3} C_O^{\text{int}} C_{\text{Cr}}^{\text{GB}}, \quad (17)$$

$$J_r^{\text{Fe}_3\text{O}_4} = k_r^{\text{Fe}_3\text{O}_4} (C_O^{\text{int}} - C_O^{\text{th}}) C_{\text{Fe}}^{\text{GB}}, \quad (18)$$

$$J_r^{\text{FeCr}_2\text{O}_4} = k_r^{\text{FeCr}_2\text{O}_4} C_O^{\text{int}} \left(\frac{1}{C_{\text{Fe}}^{\text{GB}}} + \frac{2}{C_{\text{Cr}}^{\text{GB}}} \right)^{-1}, \quad (19)$$

$$J_r^{\text{SiO}_2} = k_r^{\text{SiO}_2} C_O^{\text{int}} C_{\text{Si}}^{\text{GB}}, \quad (20)$$

where k_r^k is the interfacial rate constant of phase k and C_O^{th} is the threshold oxygen concentration for magnetite formation. Their sum gives the total interfacial consumption,

$$J_r = J_r^{\text{Cr}_2\text{O}_3} + J_r^{\text{Fe}_3\text{O}_4} + J_r^{\text{FeCr}_2\text{O}_4} + J_r^{\text{SiO}_2}. \quad (21)$$

Equations (16) and (21) constitute the quasi-steady balance between oxygen supply and oxygen consumption,

$$D_O^{\text{eff}} \frac{C_O - C_O^{\text{int}}}{\sum_k L_k} = \sum_k J_r^k(C_O^{\text{int}}), \quad (22)$$

which is solved at every position for the single unknown C_O^{int} ; the supply term decreases and the consumption term increases monotonically with C_O^{int} , so the root is unique. The corresponding rates of metal removal follow from the stoichiometry of each phase,

$$J_r^{\text{Cr}} = \frac{2}{3} J_r^{\text{Cr}_2\text{O}_3} + \frac{2}{4} J_r^{\text{FeCr}_2\text{O}_4}, \quad (23)$$

$$J_r^{\text{Fe}} = \frac{3}{4} J_r^{\text{Fe}_3\text{O}_4} + \frac{1}{4} J_r^{\text{FeCr}_2\text{O}_4}, \quad (24)$$

$$J_r^{\text{Si}} = \frac{1}{2} J_r^{\text{SiO}_2}. \quad (25)$$

2.3 Coupling Strategy and Simulation Protocol

2.3.1 Lattice Velocity and Metal Consumption

The interfacial reactions remove metal atoms only at the grain-boundary plane. Because the metal lattice is incompressible and the boundary is a perfect defect sink, this removal creates no source in the interior: the entire lattice translates towards the interface at the velocity set by the interfacial consumption. Denoting by

$$v_{\text{ox}}(y, t) = J_r^{\text{Cr}} + J_r^{\text{Fe}} + J_r^{\text{Si}} \quad (26)$$

the recession rate of the metal/oxide interface, the total lattice velocity in the metal domain is

$$\mathbf{v}_{\text{lat}}(x, y, t) = \mathbf{v}_{\text{lat}}^{\text{RIS}}(x, y, t) + v_{\text{ox}}(y, t) \hat{\mathbf{e}}_x. \quad (27)$$

The two contributions differ in character. The segregation term is local: it is determined pointwise by the defect fluxes at (x, y) . The oxidation term

is not; it is evaluated at the boundary and is independent of x , so a single value applies across the whole domain at a given y . Segregation therefore redistributes the lattice internally, whereas oxidation translates it rigidly so that the total flux of alloying element i entering Eq. (1) is

$$\mathbf{J}_i = \mathbf{J}_i^V + \mathbf{J}_i^I + C_i \mathbf{v}_{\text{lat}}, \quad (28)$$

subject to the interfacial condition

$$\mathbf{J}_i \cdot \hat{\mathbf{e}}_x|_{x=0} = J_r^i, \quad (29)$$

which removes from the boundary plane the atoms incorporated into the growing oxide.

Equations (28) and (29) collect the three ways in which an alloying element can move. The first two terms are the diffusive fluxes carried by the two defect mechanisms, Eqs. (4) and (5); the third is the drift of the entire lattice relative to the laboratory frame; the interfacial consumption enters not as a bulk term but as an outgoing flux on the boundary plane.

The two contributions to $\mathbf{v}_{\text{lat}}$ have different origins but act identically on every species. $\mathbf{v}_{\text{lat}}^{\text{RIS}}$ arises because the vacancy and interstitial fluxes are unequal, so lattice sites are created and destroyed at different rates; v_{ox} arises because oxidation consumes metal atoms from the interfacial layer, so the lattice is displaced towards the advancing front. Because the drift term $C_i \mathbf{v}_{\text{lat}}$ is proportional to the local concentration and carries no species selectivity, it does not by itself alter the relative composition; selectivity enters only through the mechanism-specific diffusivities in Eqs. (6)–(9) and through the stoichiometric sinks in Eqs. (23)–(25).

2.3.2 Sequential Irradiation–Oxidation Protocol

The reference experiments irradiate the specimen first and expose it to the primary water environment afterwards. The model reproduces this protocol as two consecutive stages that share the same governing equations and differ only in two switches: the dose rate K_0 and the oxygen boundary condition at the grain-boundary mouth.

In the irradiation stage the specimen is irradiated to the target dose with no oxygen present, $K_0 > 0$ and $C_O \equiv 0$, so that Eqs. (1)–(12) govern the evolution alone and all interfacial reaction rates vanish identically. The stage is integrated to $t_{\text{irr}} = \Phi/K_0$, where Φ is the nominal dose in dpa, and yields the segregated composition profiles $C_i(x, y, t_{\text{irr}})$.

The oxidation stage continues from the final state of the irradiation stage: all concentration fields, including the point-defect populations, are inherited unchanged,

$$(C_i, C_V, C_I)|_{t=0+} = (C_i, C_V, C_I)|_{t=t_{\text{irr}}}, \quad L_k|_{t=0+} = 0, \quad (30)$$

and the exposure clock is restarted from zero. The dose rate is then set to zero, $K_0 = 0$, so that the inherited defect supersaturation decays by recombination according to Eqs. (2) and (3) without further production.

Simultaneously, the oxygen boundary condition at the mouth of the boundary is switched on. Oxygen ingress is limited by the surface film that grows on the free surface, whose resistance increases as the film thickens; this is imposed as a Robin condition with a time-decaying transfer coefficient,

$$J_O|_{y=0} = h(t) (C_O^w - C_O|_{y=0}), \quad h(t) = \frac{k_s}{\sqrt{t} + \sqrt{t_0}}, \quad (31)$$

where C_O^w is the oxygen concentration of the bulk water, k_s is the surface transfer coefficient, and t_0 regularizes the initial instant. The $t^{-1/2}$ decay reproduces the parabolic thickening of the surface film, so that the cumulative oxygen admitted through the mouth grows as $t^{1/2}$. The deep end of the boundary is closed, $J_O|_{y=L_{\text{GB}}} = 0$.

Because no point defects are produced once $K_0 = 0$, the inherited supersaturation decays on the recombination timescale, which is short compared with the exposure time; segregation therefore effectively ceases early in the exposure stage, and the composition field that the oxidation model sees is the one established during irradiation. The sequential protocol thus isolates the compositional pathway: any dose dependence of the predicted oxidation depth must be carried by the radiation-induced segregation profile inherited through Eq. (30), since neither radiolysis nor sustained irradiation-assisted transport is present.

2.3.3 Fully Coupled Irradiation–Oxidation Framework

The sequential protocol of Sec. 2.3.2 is a special case of the governing equations rather than a separate model: it is obtained by switching K_0 and the mouth boundary condition on and off in turn. Leaving both switches on throughout, $K_0 > 0$ together with the Robin condition Eq. (31), gives the

fully coupled problem in which segregation and oxidation proceed simultaneously, as they do in reactor service. No additional equations or parameters are required; Eqs. (1)–(12) are integrated as a single stage.

In this mode the two processes feed back on each other. Segregation continues to modify the boundary composition while the oxidation front advances, so the composition that the interfacial reactions see evolves in time rather than being fixed at its end-of-irradiation value; conversely, the metal consumed at the front and the associated lattice drift, Eq. (26), alter the concentration field on which the segregation fluxes act. Predictions obtained in this mode are reported in Sec. 3.5; because no experiment in the present dataset imposes concurrent irradiation and exposure, this mode is used only for prediction and takes no part in the calibration of Sec. 2.4.

2.4 Model Parameterization and Calibration

The two components are parameterized and calibrated separately and against separate data. The RIS model is calibrated first; the oxidation model is then calibrated with the RIS model related parameters fixed.

2.4.1 RIS Parameterization

The RIS model requires the mechanism-specific diffusion coefficients D_i^V and D_i^I of the four alloying elements, the correlation factors f_0^V and f_0^I , the recombination coefficient R , the damage efficiency ϵ , and the dose rate K_0 . The dose rate and irradiation temperature are those of the reference experiment. The correlation factors are taken from their crystallographic values for the fcc lattice. The damage efficiency accounts for the fraction of displacements that survive in-cascade recombination and escape as freely migrating defects; its value is that reported for proton irradiation. The vacancy and interstitial migration energies and pre-exponential factors are taken from the literature and adjusted within their reported ranges to reproduce the measured grain-boundary composition profiles across the dose series. Because these profiles are measured before exposure, the RIS calibration is independent of the oxidation model. The resulting values are given in Table 1.

2.4.2 Oxidation Kinetics Parameterization

The oxidation model requires the interfacial rate constants k_r^k of the four oxide phases, their oxygen diffusivities D_O^k , the surface transfer coefficient k_s of the Robin condition Eq. (31), and the magnetite threshold C_O^{th} . The oxide molar volumes and densities used in the thickness update are taken from crystallographic data and held fixed.

The model is calibrated against the dose-dependent grain-boundary oxidation depths of, measured after exposure in simulated PWR primary water at 0, 0.5, and 3 dpa. The predicted oxidation depth is defined as the distance from the boundary mouth to the deepest position at which the total oxide thickness $\sum_k L_k$ exceeds a fixed threshold L_{th} , applied identically at every dose. The free parameters are adjusted to reproduce the measured depths at all three doses simultaneously, using a single parameter set; no parameter is allowed to take different values at different doses. The resulting values are given in Table 2.

Table 1: Parameters of the radiation-induced segregation model.

Symbol	Description	Value	Source
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Table 2: Parameters of the grain-boundary oxidation model.

Symbol	Description	Value	Source
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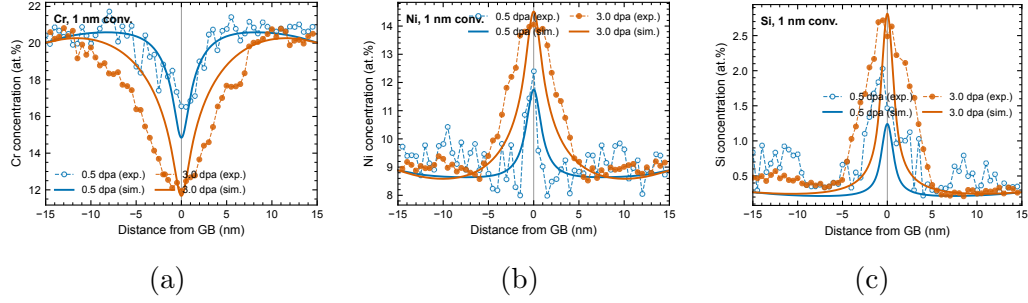


Figure 1: Calculated and measured grain-boundary composition profiles after irradiation to 0.5 and 3.0 dpa: (a) Cr, (b) Ni, (c) Si. Solid lines are calculated profiles convolved with a 1 nm Gaussian; open and filled symbols are the measurements of at 0.5 and 3.0 dpa respectively.

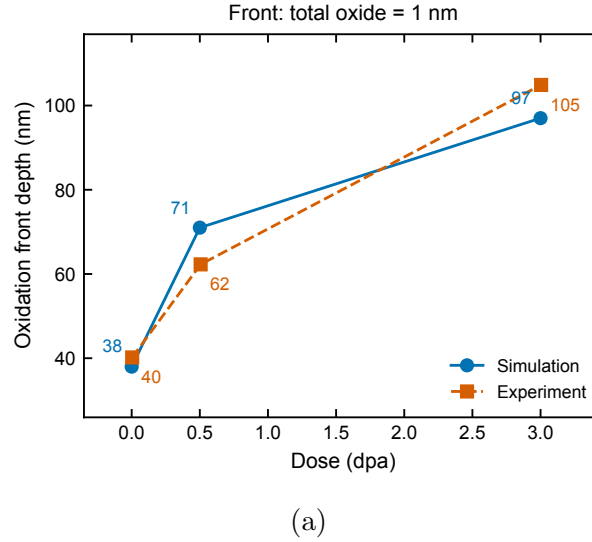


Figure 2: Oxidation front depth as a function of dose, taken at a total oxide thickness of 1 nm. Measurements from **deng** are those used to calibrate the oxidation parameters.

3 Results

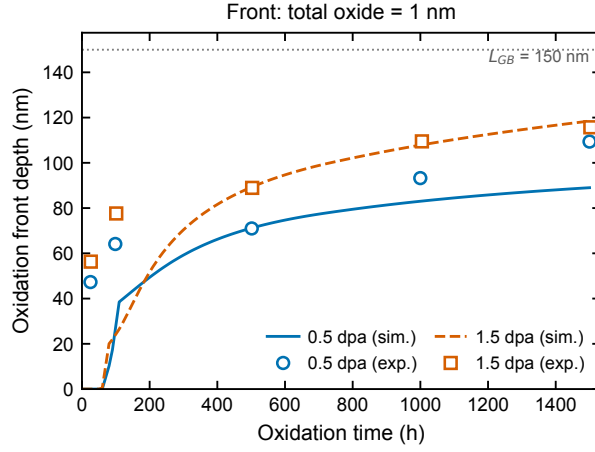
3.1 Radiation-Induced Segregation Profiles

3.2 Dose-Dependent Grain-Boundary Oxidation Kinetics

3.3 Oxide Composition and Interfacial Ni Enrichment

3.4 Predictions against Independent Experiments

3.5 Fully Coupled Predictions under Concurrent Irradiation and Oxidation



(a)

Figure 3: Predicted and measured oxidation front depth as a function of exposure time at 0.5 and 1.5 dpa, taken at a total oxide thickness of 1 nm. The parameters are those calibrated in Fig. 2; neither the 1.5 dpa dose nor the exposure times shown enter the calibration. Measurements from **deng**.